Basis Risk in USD/INR Forwards

Modelling the forward basis curve and the 31 March charge, and testing VaR models for basis risk

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Overview

This project looks at the cost of hedging dollars with USD/INR forward contracts, and in particular at the forward basis: the part of the forward premium that the interest rate gap between India and the US does not explain.

I build a daily basis curve from one to twelve months, using FBIL forward premia, FBIL MIBOR-OIS rates for the rupee and US Treasury yields for the dollar, from July 2018 to October 2026. Because the dollar leg uses Treasury yields rather than a dollar funding rate, the series is a proxy for the cross-currency basis.

I model the curve with a dynamic Nelson–Siegel state-space model, estimated by maximum likelihood with the Kalman filter on 2018–2023 data. I add a fourth factor for the 31 March financial year end, because forwards that settle after the year end carry an extra charge that a standard curve model would miss. I check this factor against FBIL's FBD April contract, with a likelihood-ratio test and with a small Monte Carlo study.

I then use the model's one-day forecasts to build Value-at-Risk and Expected Shortfall estimates, and backtest them against historical simulation and RiskMetrics on January 2024 to October 2026. I also run local projections on monthly data to see how RBI dollar purchases and changes in banking liquidity move the basis curve, and I replay three historical stress episodes on the current curve to put the risk in rupee terms.

What I find

- The basis accounts for most of the variance of daily changes in short-dated forward premia: 98% at one month, 94% at three months and 74% at twelve months. Interest rates mattered more only in 2022–24, a period that includes the rate rises by the RBI and the Fed.
- The 31 March charge can be measured every day. The model's estimate tracks the price implied by FBIL's FBD April contract (correlation 0.96, and 0.95 even when the model is not shown that price). A likelihood-ratio test strongly favours keeping the factor (LR 5,647, against at most 5.7 in simulations without it). At the rollover, the one-month basis fell in every one of the eight years (median −32 bp on the day, −69 bp the next day) while the twelve-month basis did not move.
- The charge varies a lot from year to year, from close to zero in 2024 to about 29 bp of notional in 2020. As of 6 October 2026, the charge built into forwards that span 31 March 2027 is about 6.6 bp (90% band 3.2–10.0), roughly 6.4 paise per dollar.
- **Standard VaR models understated short-dated basis risk in 2024–26. At 99%, RiskMetrics was in the Basel-style red zone at every tenor from one to seven months, and historical simulation was in the yellow zone at 11 of 12 tenors. Daily basis changes are strongly fat-tailed (fitted Student-t**

degrees of freedom of about 4 to 6), but switching to fat-tailed distributions was not enough on its own: both Student-t models were in the red zone at 95% at the shortest tenors. Filtered historical simulation on the state-space forecast (SS-FHS), which uses the distribution of past forecast errors instead of assuming one, did best at the short end: 8–9 breaches at one to three months against 14–19 for the other models, no red-zone result at any tenor, and the best Expected Shortfall test results. The normal state-space model had slightly fewer breaches in total, mostly at longer tenors, but failed the Expected Shortfall test at 7 of 12 tenors. **Its losses were lower than RiskMetrics' at one to six months, though these gains are not significant once I allow for testing 12 tenors at once.**

- **RBI dollar purchases are followed by a flatter basis curve: about −3 bp in the slope factor per USD 1 billion after two to three months, significant from month 0 to month 5.** Changes in banking liquidity show effects at single horizons only, so I treat them as suggestive.
- Replaying the March–April 2026 stress on today's curve widens the three-month basis by about 199 bp, which costs roughly ₹48.5 lakh on a USD 10 million forward.
- A small Monte Carlo study shows that the estimator recovers the year-end dynamics well, but not the long-run means or the persistence of the curve factors.

Limitations

- The dollar leg is a Treasury yield, not a funding rate, so the basis is a proxy.
- The model was estimated on 2018–23 and tested on 2024–26, but I adjusted the risk model design after seeing early test results. The test is therefore not fully independent; I will re-check it on data after October 2026.
- There are only eight year ends in the sample, and the persistence and long-run means of the factors are weakly identified.
- The year-end factor is modelled as a random walk. This works for measuring today's charge, but not for forecasting its level next March.
- The VaR is for a constant-tenor basis change, not the profit and loss of an actual forward book.
- At 95%, the best model's breaches cluster in time at mid tenors, and its gains over RiskMetrics are not significant once the 12 tenors are tested together.
- The local projections use about 95 months of data, show associations rather than causal effects, and rely on a hand-assembled RBI data panel.
- The Monte Carlo study has only 10 replications; λ and the other parameters are held fixed over the whole sample; and the data are FBIL benchmark fixings rather than traded prices.

At a glance

Key numbers

Tenor	Basis share of daily premium variance	99% breaches: RiskMetrics	99% breaches: HS	99% breaches: SS-FHS	Mar–Apr 2026 stress (bp)	Stress cost, ₹ lakh per USD 10m
1M	98.2%	18 (R)	15 (Y)	8 (G)	265	22.4
3M	93.5%	18 (R)	14 (Y)	9 (G)	199	48.5
6M	84.9%	23 (R)	16 (Y)	12 (Y)	139	66.7
12M	74.5%	16 (Y)	12 (Y)	9 (G)	81	78.0

Breaches over 666 test days (Jan 2024 – Oct 2026); expected 6.7. Basel-style zones rescaled to 666 days: G 0–10, Y 11–17, R 18+. Stress: the Mar–Apr 2026 episode's largest 10-day widening, applied to the 6 Oct 2026 curve.

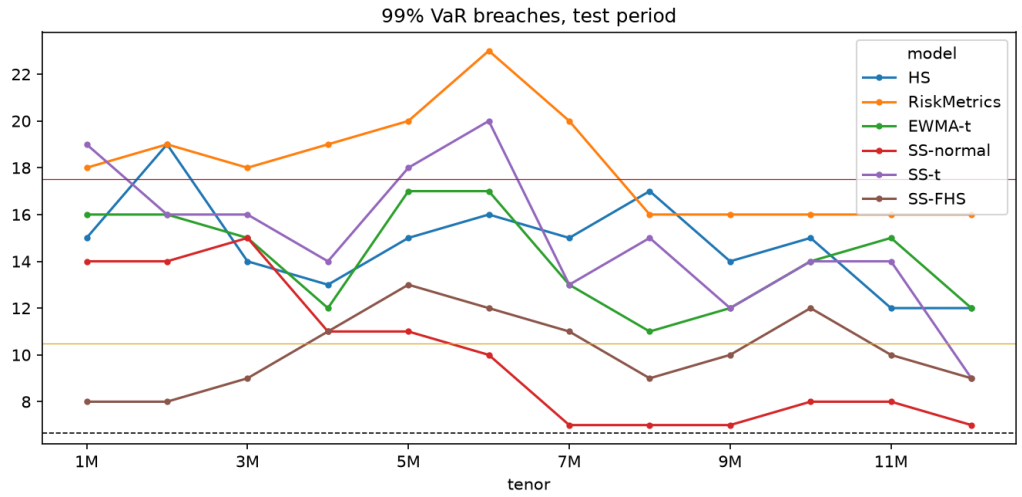


Figure S1. 99% VaR breaches by tenor, Jan 2024 – Oct 2026

Number of times each model's one-day 99% VaR was exceeded over 666 test days. Dashed line: expected number (6.7). Yellow and red lines: start of the Basel-style yellow (11) and red (18) zone

- **Short-dated forward exposure may be better monitored on the basis itself**, not only on the interest rate differential.
- **VaR models that ignore the basis curve and the year end may understate short-dated risk.** Comparing them with a curve-aware measure would show how large the gap is for a given position.
- **Exposures that span 31 March may need a separate stress test, and a daily estimate of the year-end charge could help in pricing hedges across it.**
- **Fat-tailed distributions alone may not be enough.** Basis changes are strongly fat-tailed, but replacing the normal with a Student-t did not fix the VaR. Using the distribution of past forecast errors worked better, so **filtered historical simulation** may be worth testing before simply changing the distribution.

1.Introduction

A firm that buys dollars three months forward pays the spot rate plus a forward premium. Under CIP the premium is set by the interest rate differential, so a treasurer could in principle forecast hedging cost from rupee and dollar rates. Since 2008 CIP has failed persistently in major currencies (Du, Tepper and Verdelhan, 2018; Borio et al., 2016). The gap is linked to balance-sheet constraints on banks and to the strength of the dollar (Avdjiev et al., 2019), and it tends to widen around reporting dates such as quarter ends (Du, Tepper and Verdelhan, 2018).

In India the main reporting date is 31 March, the end of the financial year. Forward contracts that settle after the year end may carry an extra charge for crossing it. This matters to anyone hedging over March, but the charge is not quoted as a separate price, so it has to be inferred from the curve.

The project asks three questions. How much of the movement in short-dated forward premia comes from the basis rather than from interest rates? Can the year-end charge be measured and tracked over time? Do standard risk models such as historical simulation and RiskMetrics measure basis risk correctly, and does a model that takes the calendar into account do better?

The approach is to treat the basis at twelve tenors as a curve, describe it with level, slope and curvature factors in the dynamic Nelson–Siegel form (Nelson and Siegel, 1987; Diebold and Li, 2006; Diebold, Rudebusch and Aruoba, 2006), add a fourth factor for the year-end charge, and estimate the system by maximum likelihood with the Kalman filter. The model's one-day forecasts are then used to build Value-at-Risk (VaR) and Expected Shortfall (ES) forecasts, which are backtested against standard benchmarks.

2. Data

The sample covers 1,977 trading days from 30 July 2018 to 6 October 2026. The first 1,311 days (to 31 December 2023) are used for estimation; the last 666 days (1 January 2024 onward) are kept for testing the risk models.

Table 1. Data sources

Series	Source	Use
USD/INR forward premia, 1M–12M and FBD April	FBIL	Forward premium; year-end contract
MIBOR-OIS curve (1M, 2M, 3M, 6M, 9M, 1Y)	FBIL	Rupee interest rate
Treasury bill curve	FBIL	Alternative rupee rate (check)
Treasury par yield curve, to Dec 2023	US Treasury	Dollar interest rate
DGS1MO, DGS3MO, DGS6MO, DGS1, from Jan 2024	FRED	Dollar interest rate
Treasury bill rates, 2024–26	US Treasury	Check on dollar rate
Net spot purchases, forward position, liquidity	RBI	Local projections (monthly)
Broad dollar index (DTWEXBGS)	FRED	Control in local projections

FBIL: Financial Benchmarks India Pvt Ltd. FBD April is FBIL's forward for settlement on the first business day of April.

Rupee rates are MIBOR-OIS rates and dollar rates are US Treasury constant-maturity yields, both interpolated linearly in days to each contract's settlement date. US rates are taken from the previous US business day; the largest gap in the sample is four calendar days. Table 2 summarises the basis.

Table 2. CIP basis by tenor, full sample (bp, annualised)

Tenor	Mean	SD	Min	Max	SD of daily change
1M	45.8	84.6	-143.6	590.3	28.1
2M	48.6	71.1	-105.2	459.6	17.3
3M	52.0	64.2	-89.9	381.8	13.2
4M	53.0	58.7	-78.5	321.8	11.0
5M	53.5	54.8	-69.4	281.9	10.0
6M	53.9	52.0	-55.7	251.5	8.7
7M	53.4	49.8	-49.4	236.4	8.5
8M	52.3	47.6	-57.6	222.7	7.9
9M	51.3	45.4	-54.7	210.4	8.0
10M	50.5	43.6	-49.6	202.5	7.2
11M	49.5	42.1	-49.2	198.3	6.8
12M	48.7	41.0	-53.9	194.4	7.0

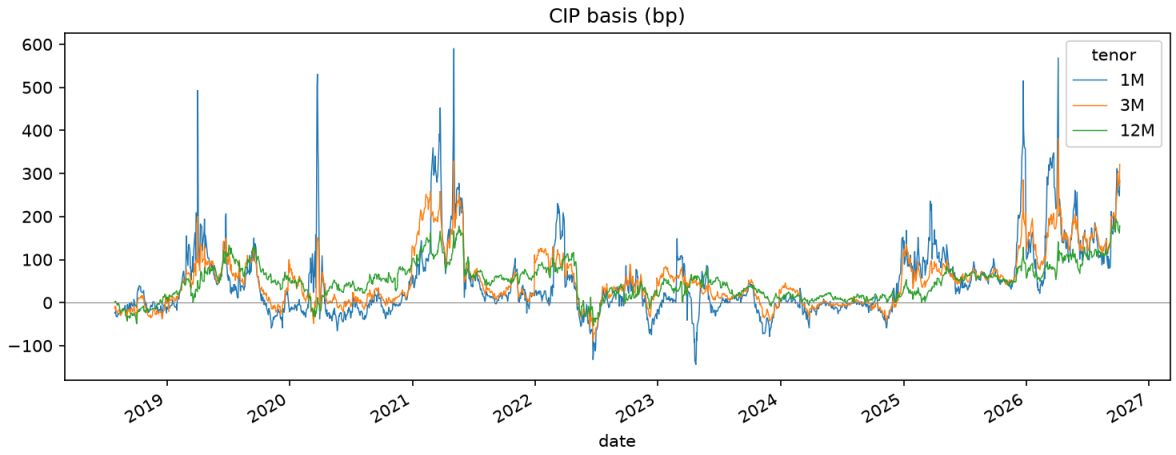


Figure 1. CIP basis at 1M, 3M and 12M (bp)

Basis = FBIL forward premium minus the CIP-implied premium, in annualised basis points. A positive value means dollars cost more forward than the interest rate differential implies.

3. Method in brief

Full details are in Appendices A to E.

Basis. For each contract, the CIP-implied premium is computed from the rupee and dollar rates for the contract's exact number of days, using simple interest on an actual/365 basis. The basis is the FBIL premium minus the CIP-implied premium, in annualised basis points. Because the dollar rate is a

Treasury yield rather than a dollar funding rate, this is a proxy for the cross-currency basis. A positive basis means dollars cost more forward than interest rates alone would imply.

Model. The basis at the twelve tenors is described by three Nelson–Siegel factors (level, slope, curvature), each following an AR(1) process, with λ fixed at 0.30 (per month). A fourth factor measures the year-end charge in basis points of notional. It loads only on contracts that settle after the coming year end, moves as a random walk within each financial year, and restarts after the rollover. FBIL's FBD April contract enters as an extra observation. Tenor-level deviations from the curve follow AR(1) processes that are correlated across neighbouring tenors. Parameters are estimated by exact maximum likelihood using the Kalman filter in statsmodels.

Risk models. The risk variable is the one-day change in the basis at each tenor. Six one-day VaR and ES models are compared: historical simulation (HS), RiskMetrics (EWMA, normal), an EWMA model with Student-t tails and a deterministic year-end adjustment (EWMA-t), and three models built on the state-space one-step forecast: normal (SS-normal), Student-t with an EWMA scale (SS-t), and filtered historical simulation (SS-FHS). Forecasts are evaluated with the Kupiec and Christoffersen tests, the Basel traffic light, the Du–Escanciano ES test and Diebold–Mariano comparisons of tick and FZ0 losses.

Other analyses. A covariance decomposition splits the variance of premium changes into rate and basis parts. A likelihood-ratio test and a Monte Carlo study check the year-end factor and the estimator. An event study looks at the basis around each rollover. Monthly local projections relate the smoothed factors to RBI intervention and liquidity. Historical stress episodes are replayed on the latest curve.

4. Results

4.1 Where hedging-cost risk comes from

Table 3 shows the share of the variance of daily forward premium changes that comes from the basis. Across the whole sample, the basis accounts for almost all of the variance at 1M and three quarters at 12M. This is a covariance share: it says how much of the variance of premium changes moves with the basis, not that the basis caused each daily move. The share falls with tenor, mainly because the basis is far more volatile at the short end: its daily changes have a standard deviation of 28 bp at 1M against 7 bp at 12M (Table 2). In 2022–24, a period that includes the rate rises by the RBI and the Federal Reserve, the rate share rose and the basis share at 12M fell to 45.1%. In every other period the basis accounted for more than three quarters of the variance at all tenors.

Table 3. Basis share of the variance of forward premium changes (%)

Tenor	2018–19	2020–21	2022–24	2025–26	Full (daily)	Full (weekly)	Full (month-end)
1M	99.7	98.6	92.9	98.1	98.2	93.4	89.8
3M	98.5	96.0	75.8	95.2	93.5	89.6	82.2
6M	95.1	92.4	59.7	88.6	84.9	80.2	72.7
9M	89.0	94.1	49.0	82.0	80.7	71.4	59.8
12M	85.7	90.1	45.1	79.9	74.5	66.9	54.8

Share = Cov(Δbasis, Δpremium) / Var(Δpremium). The rate share is the remainder. All twelve tenors are reported in the run output.

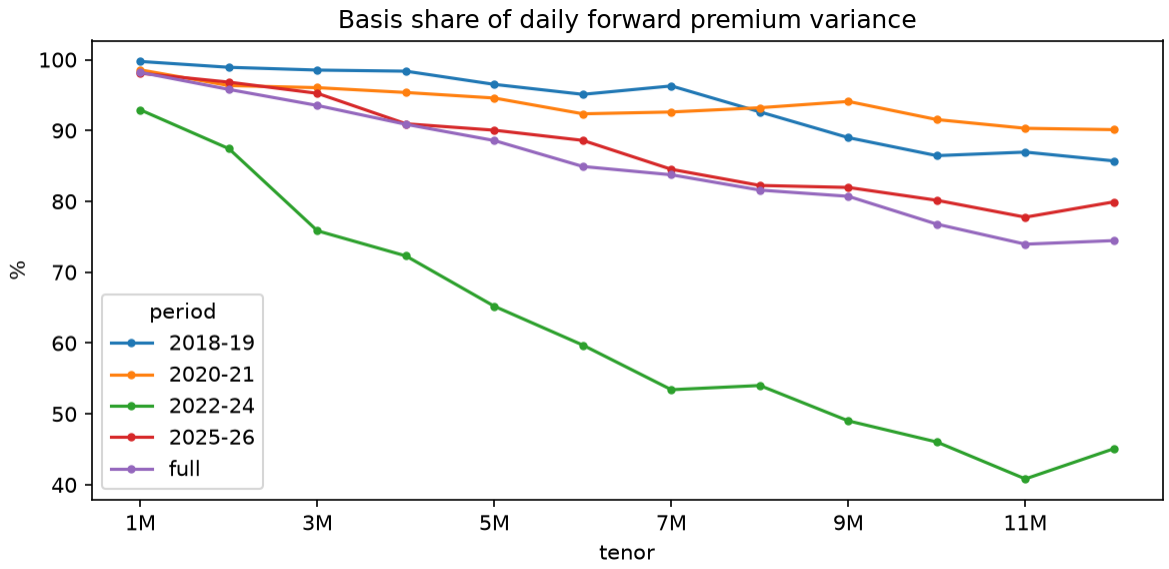


Figure 2. Basis share of daily forward premium variance by tenor and period

Each line shows, for one period, the share of the variance of daily forward premium changes that comes from the basis (Cov(Δbasis, Δpremium) / Var(Δpremium)). The rest comes from the interest rate differential.

4.2 Model estimates and fit

All five fits converged. The main model's log-likelihood is -49244.9 over the training sample, and it was identical in three separate runs. The three curve factors are highly persistent, with daily AR(1) coefficients between 0.974 and 0.977, which correspond to half-lives of roughly 26 to 29 trading days. Selected estimates are in Table 4; the full table is in Appendix C.

Table 4. Selected parameter estimates

Parameter	Meaning	Estimate	Std. error
ϕ_L, ϕ_S, ϕ_C	Daily persistence of level, slope, curvature	0.974, 0.977, 0.975	0.005, 0.007, 0.006
σ_w	Daily change in year-end charge (bp of notional)	2.66	0.09
μ_D	Charge at the start of each season (bp)	0.12	1.30
ρ	Persistence of tenor deviations	0.581	0.008
κ	Correlation decay between tenors, per month	0.472	0.016
σ_1M, σ_3M, σ_12M	Size of tenor deviations (bp)	17.2, 6.2, 4.5	0.47, 0.18, 0.10

Standard errors from the numerical Hessian at the estimates.

Table 5 compares the main model with restricted and alternative versions. Holding the year-end charge fixed within each season lowers the log-likelihood by 1290 points, so the charge does move within the year. The fit is not very sensitive to λ: λ = 0.4 fits slightly better (by 1.6 points) and λ = 0.2 slightly worse, with similar parameter estimates in both cases.

Table 5. Model comparison (training sample)

Model	Log-likelihood	Difference from main
Main model (λ = 0.30)	-49244.9	0.0
Charge fixed within each season	-50535.0	-1290.1
λ = 0.2	-49255.5	-10.6
λ = 0.4	-49243.3	1.6
Without the FBD April contract	-46333.3	not comparable

The model without the FBD April contract is fitted to different data, so its log-likelihood cannot be compared with the others.

The model fits the curve closely beyond one month. Root mean squared errors are between 1 and 6 bp for 2M–12M, against daily moves of 7 to 17 bp. At 1M the error is 15.2 bp, against daily moves of 28.1 bp; the 1M tenor is the noisiest point on the curve. Figure 3 shows the smoothed factors.

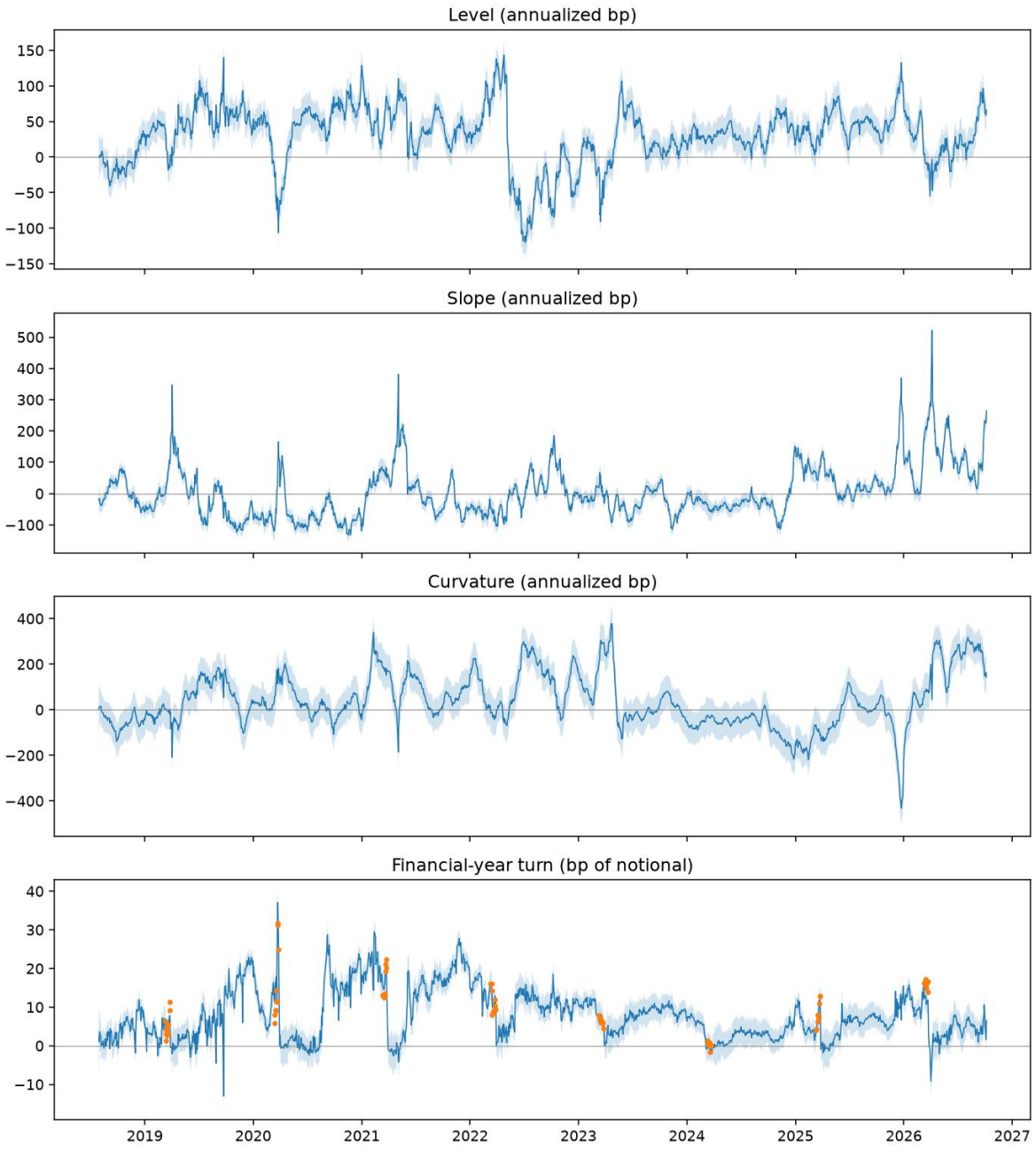


Figure 3. Smoothed factors with 90% bands

Level, slope and curvature in annualised bp; year-end charge in bp of notional. Shaded areas: 90% bands. Orange points in the bottom panel: year-end charge implied by FBIL's FBD April contract.

4.3 The financial year-end charge

The FBD April contract settles on the first business day of April. In the last few weeks before the year end, I use its basis, minus an estimate of ordinary carry taken from 1M contracts that do not span the year end, as a proxy for the year-end charge. The model is checked against this price on the 82 days on which it is available (Table 6). The version estimated without the FBD contract is the stronger test, because it has to infer the charge from the shape of the curve alone.

Table 6. Model year-end charge against the FBD April price

Model	Days	Correlation	RMSE (bp)	Mean gap (bp)
Main model (uses FBD April)	82	0.958	2.69	0.95
Model without FBD April	82	0.954	3.13	1.18

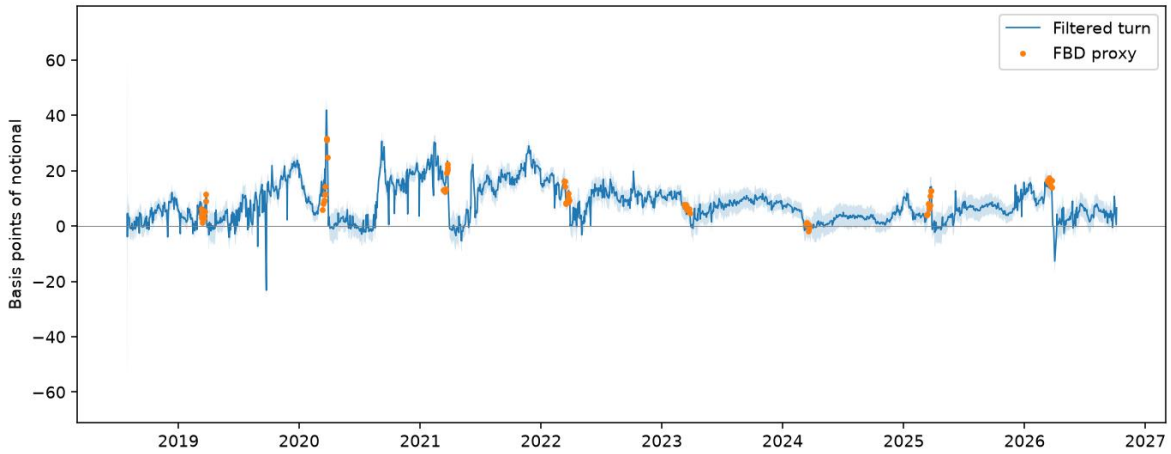


Figure 4. Filtered year-end charge and the FBD April price (bp of notional)

Blue line: filtered estimate of the year-end charge, using only data up to each day, with 90% band. Orange points: charge implied by the FBD April contract in the weeks before each 31 March. The band is wide at the start of the sample while the filter initialises.

Table 7 reports the charge each year. It peaked at about 29 bp of notional in March 2020 and was close to zero in 2024. On the last full day before the rollover in 2026 it stood at 14.4 bp, or about 13.6 paise per dollar.

Table 7. Year-end charge by financial year (bp of notional)

Year end	End-February	Last full day	90% band	Paise per USD	Observed FBD median
2019	7.6	11.0	7.3 to 14.8	7.6	4.3
2020	6.7	28.7	25.1 to 32.4	21.6	11.5
2021	18.0	22.4	18.7 to 26.1	16.3	13.4
2022	9.4	11.6	8.1 to 15.2	8.9	9.4
2023	9.3	4.6	1.0 to 8.2	3.8	6.2
2024	3.5	-0.7	-4.5 to 3.1	-0.6	0.4
2025	4.5	14.3	10.5 to 18.1	12.3	7.4
2026	8.2	14.4	10.7 to 18.1	13.6	16.2

Model values are filtered estimates, using only information available on that day.

As of 6 October 2026, the charge built into forwards that span 31 March 2027 is estimated at 6.6 bp of notional (90% band 3.2 to 10.0 bp), or about 6.4 paise per dollar. This is the charge priced today. The charge observed next March could be quite different (Section 5).

Is the factor needed? The likelihood-ratio statistic for removing the year-end factor is 5,647. Because the restricted model sets a variance to zero, the usual χ^2 distribution is not a reliable guide (Self and Liang, 1987), so the null distribution was simulated. In ten simulations from a model with no year-end charge, the LR ranged from 0 to 5.7, and none exceeded the $\chi^2(3)$ 5% critical value of 7.8. With only ten replications the Monte Carlo p-value cannot fall below 1/11 ≈ 0.09, so I report the comparison directly: the observed statistic is far above the largest simulated value.

Behaviour at the rollover. Figure 5 shows the median daily change in the basis around each rollover. Day 0 is the last trading day of the old season. The 1M basis falls on day 0 in 8 of 8 years (median -32.0 bp) and again on day 1 in 7 of 8 years (median -68.9 bp). The 3M basis falls by less, and the 12M basis does not move (median -0.0 and 0.6 bp). This is consistent with the charge sitting in contracts that span the year end and dropping out once they no longer do. The basis also rises in the days before the year end (days −3 and −2).

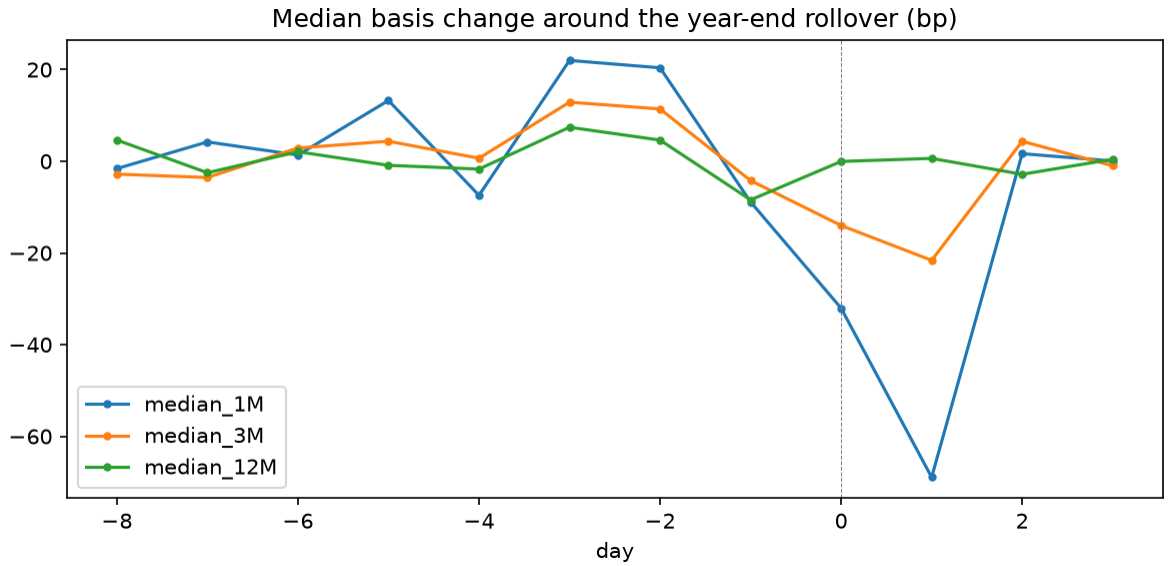


Figure 5. Median daily change in the basis around the year-end rollover (bp)

Median across the eight year ends of the daily change in the 1M, 3M and 12M basis (labelled median_1M, median_3M, median_12M), by trading day relative to the rollover. Day 0 is the last trading day of the old season.

4.4 RBI intervention and liquidity

Local projections (Jordà, 2005) relate the monthly average of the smoothed level and slope factors h months ahead to RBI net spot purchases and to the change in net liquidity injection. Controls are two lags of all variables (including the change in RBI's net forward position and the broad dollar index) and month dummies. The sample is 95 months; standard errors are Newey–West (HAC).

A USD 1 billion increase in net spot purchases is followed by a lower slope: -1.5 bp on impact, about -3.1 bp after two and three months, and close to zero by month six. The 90% band excludes zero from month 0 to month 5. A lower slope means the short end of the basis curve falls relative to the long end. The other responses are significant at one horizon at most. With 28 coefficients at the 90% level, about three

would be expected to exclude zero by chance, so these should be treated as suggestive. All estimates are associations conditional on the controls; RBI intervention itself responds to market conditions.

Table 8. Response of the slope factor to RBI net spot purchases (bp per USD 1 billion)

Horizon (months)	Response	HAC s.e.	90% band
0	-1.46	0.74	-2.68 to -0.24
1	-2.07	1.00	-3.71 to -0.42
2	-3.14	1.09	-4.93 to -1.35
3	-3.14	0.98	-4.75 to -1.53
4	-2.95	0.86	-4.36 to -1.54
5	-1.14	0.68	-2.26 to -0.03
6	-0.68	1.08	-2.45 to 1.09

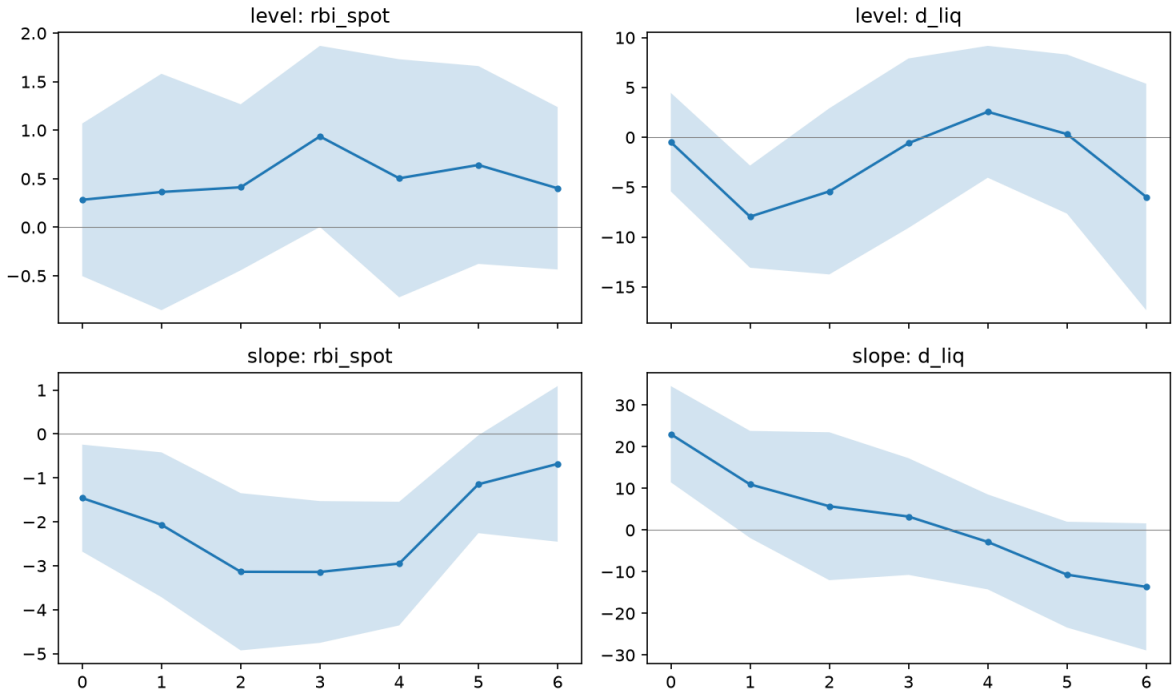


Figure 6. Local projections: level and slope responses with 90% bands

Line: estimated change in the factor (bp) h months after a one-unit change in the driver. Shaded area: 90% band using HAC standard errors. Top row: level factor; bottom row: slope factor. rbi_spot: RBI net spot purchases (USD billion). d_liq: change in net liquidity injection (₹ lakh crore).

4.5 Measuring basis risk

Each model produces a one-day 99% and 95% VaR for an increase in the basis at each tenor, which is the risk faced by a firm that is due to buy dollars forward. Over 666 test days, a correct 99% VaR should be breached about 6.7 times. With **Basel-style traffic-light cut-offs** rescaled to 666 days, 0–10 breaches is green, 11–17 yellow and 18 or more red.

Table 9. Number of 99% VaR breaches in the test period (Basel-style zone)

Tenor	HS	RiskMetrics	EWMA-t	SS-normal	SS-t	SS-FHS
1M	15 (Y)	18 (R)	16 (Y)	14 (Y)	19 (R)	8 (G)
2M	19 (R)	19 (R)	16 (Y)	14 (Y)	16 (Y)	8 (G)
3M	14 (Y)	18 (R)	15 (Y)	15 (Y)	16 (Y)	9 (G)
4M	13 (Y)	19 (R)	12 (Y)	11 (Y)	14 (Y)	11 (Y)
5M	15 (Y)	20 (R)	17 (Y)	11 (Y)	18 (R)	13 (Y)
6M	16 (Y)	23 (R)	17 (Y)	10 (G)	20 (R)	12 (Y)
7M	15 (Y)	20 (R)	13 (Y)	7 (G)	13 (Y)	11 (Y)
8M	17 (Y)	16 (Y)	11 (Y)	7 (G)	15 (Y)	9 (G)
9M	14 (Y)	16 (Y)	12 (Y)	7 (G)	12 (Y)	10 (G)
10M	15 (Y)	16 (Y)	14 (Y)	8 (G)	14 (Y)	12 (Y)
11M	12 (Y)	16 (Y)	15 (Y)	8 (G)	14 (Y)	10 (G)
12M	12 (Y)	16 (Y)	12 (Y)	7 (G)	9 (G)	9 (G)

Expected breaches: 6.7. G = green (0–10), Y = yellow (11–17), R = red (18+). Zone cut-offs are the binomial 95% and 99.99% points for 666 days, the same rule the Basel framework uses for 250 days.

- **RiskMetrics understated risk at the short end.** At 99% it was in the red zone at every tenor from 1M to 7M and never green. Its ES test rejects at every tenor ($p < 0.001$).
- **Historical simulation did little better.** It was yellow at 11 tenors and red at 2M.
- **SS-FHS did best at the short end.** It had 8–9 breaches at 1M–3M against 14–19 for the other models, was never in the red zone, and its ES test does not reject at 10 of 12 tenors (the exceptions are 5M and 7M). Two other models also stayed out of the red zone at 99%: EWMA-t, and SS-normal, which had slightly fewer breaches in total (119 against 122 for SS-FHS) because of low counts at 7M–12M. But SS-normal failed the ES test at 7 of 12 tenors, and at 95% it was too conservative: 278 breaches across all tenors against about 400 expected.
- **Student-t tails did not help.** SS-t, which uses the same forecast with Student-t tails, was red at 1M, 5M and 6M at 99% and red at 1M–3M at 95%. With fitted degrees of freedom of about 4 to 6, the t distribution puts its 95% quantile too low. **SS-FHS** does not assume a tail shape; it uses the distribution of past forecast errors.

Diebold–Mariano tests on the FZ0 loss (Patton, Ziegel and Chen, 2019), which scores VaR and ES jointly, show where the improvement is. At 99%, SS-FHS has lower loss than RiskMetrics at 1M through 6M, with individual p-values between 0.015 and 0.038, but not at 7M–12M. Because twelve tenors are tested, I also apply a Benjamini–Hochberg correction; the adjusted p-values are then about 0.08, so I would not rely on the result at any single tenor. Against HS the difference is individually significant only at 1M and 2M (Appendix D). **The gain is concentrated at the short end, which is where the year-end charge and the largest moves are.**

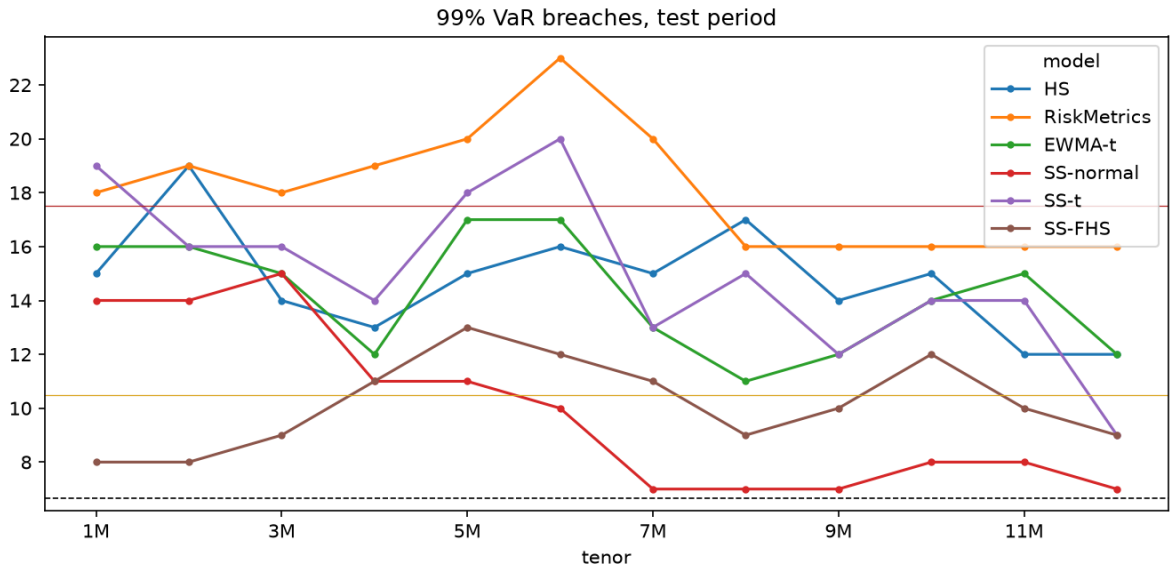


Figure 7. 99% VaR breaches by tenor and model, test period

Number of times each model's one-day 99% VaR was exceeded over 666 test days. Dashed line: expected number (6.7). Yellow and red lines: start of the Basel-style yellow (11) and red (18) zones.



Figure 8. Daily basis changes and 99% VaR: RiskMetrics and SS-FHS

Grey: daily change in the basis (bp). Orange: RiskMetrics 99% VaR (labelled EWMA-normal). Blue: SS-FHS 99% VaR. Test period only. A breach is a day on which the grey line rises above a VaR line.

Figure 8 shows the two models through the test period. In the December 2025 and March–April 2026 episodes, RiskMetrics VaR rises mostly after the large moves, while SS-FHS rises further and faster. One possible reason is that the state-space forecast uses information from the whole curve; another is that the empirical error distribution has fatter tails than the normal. I have not separated the two. SS-FHS also has a weakness: at 95%, its breaches cluster in time at 4M–10M (Christoffersen test $p < 0.05$). The number of breaches is right, but they come in groups. The risk measured here is the one-day change in a constant-tenor basis, not the P&L of an actual forward book, and the SS-FHS result comes from a test period that was not fully independent (Section 5).

4.6 Stress scenarios

Each historical episode is summarised by the 10-day window with the largest rise in the 3M basis. The changes in the level, slope and curvature factors over that window are applied to the curve as of 6 October 2026, and converted into rupees using current contract lengths and the spot rate. The year-end charge is not included.

Table 10. Historical stress scenarios applied to the October 2026 curve

Tenor	Covid 2020 (bp)	Dec 2025 (bp)	Mar-Apr 2026 (bp)	Covid 2020 (₹ lakh)	Dec 2025 (₹ lakh)	Mar-Apr 2026 (₹ lakh)
1M	168.5	205.9	265.3	14.2	17.4	22.4
3M	130.4	135.6	199.4	31.7	33.0	48.5
6M	82.2	84.3	138.7	39.5	40.5	66.7
12M	18.3	54.3	80.9	17.7	52.3	78.0

₹ lakh per USD 10 million notional. Windows: Covid 2020, 2020-03-11 to 2020-03-26; Dec 2025, 2025-12-09 to 2025-12-23; Mar–Apr 2026, 2026-03-16 to 2026-04-06.

The March–April 2026 episode is the most severe at every tenor. Although the change in bp falls with tenor, the rupee cost rises with tenor up to six months in all three episodes, because the annualised change applies over a longer contract. In the two recent episodes it keeps rising to 12M.

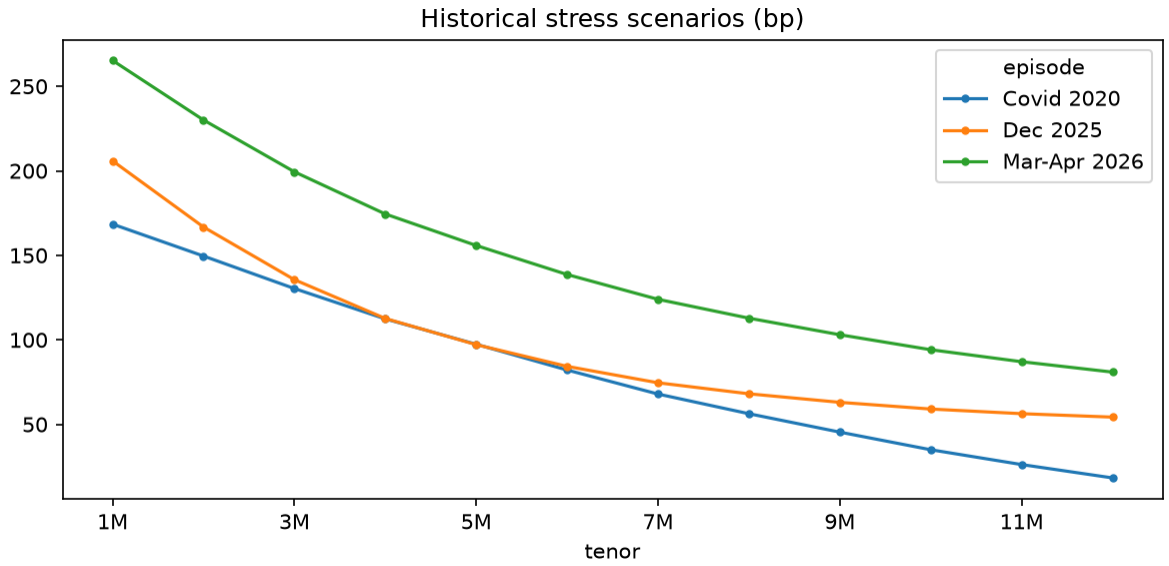


Figure 9. Historical stress scenarios across the curve (bp)

Change in the basis at each tenor when the factor moves from each episode's worst 10-day window are applied to the curve of 6 October 2026. Annualised bp; the year-end charge is excluded.

5. Limitations

1. **The test period is not fully independent.** Parameters were estimated on 2018–23 only, and each forecast uses only past data. But the risk model design (for example, using a separate volatility scale for each tenor) was adjusted after looking at 2024–26 results. When test data help choose a model, they tend to flatter it. RiskMetrics and historical simulation were not tuned at all, so their results are unaffected. The SS-FHS result should be re-checked on data after October 2026.
2. **Risk factor, not book P&L.** The VaR is for the one-day change in a constant-tenor basis. A real forward book rolls down the curve, has different notionals and may be partly hedged, so its P&L risk will differ.

3. The FZ0 gains of SS-FHS over RiskMetrics at 1M–6M have p-values between 0.015 and 0.038 individually, but after a Benjamini–Hochberg correction across the 12 tenors the adjusted p-values are about 0.08.
4. **The year-end factor is not a forecasting model.** As a random walk with a daily standard deviation of 2.66 bp, it implies a 90% band of roughly ±47 bp for the charge five months ahead, wider than any charge observed. The model measures the current charge well, but a mean-reverting or seasonal specification would be needed to forecast it.
5. **The dollar leg uses Treasury yields, not SOFR OIS.** CIP is usually measured with rates at which banks can borrow and lend, such as OIS rates (Du, Tepper and Verdelhan, 2018). Part of the basis measured here is therefore the gap between Treasury yields and dollar OIS, which I have not measured.
6. **Few year ends.** The sample has eight year ends, five of them in the estimation period. The starting level of the charge (μ_D) and its jump at the restart (σ_ξ, estimated at zero) are poorly determined.
7. **Small Monte Carlo study.** Ten replications are enough to show that the estimator recovers the year-end dynamics, but not to give precise p-values or coverage rates. One of the 20 simulation fits did not converge.
8. **Fixed parameters.** The model assumes constant parameters over 2018–26, although the 2022–24 period behaved differently (Table 3). λ is fixed rather than estimated.
9. **Interpolation and curve points.** Rates are interpolated linearly between curve points, and the US curve after 2023 uses only the 1M, 3M, 6M and 1Y points.
10. SS-FHS breaches at 95% cluster at mid tenors, and SS-FHS fails the ES test at 5M and 7M.
11. **Local projections.** The sample is short (about 95 months), the estimates are associations rather than causal effects, and many coefficients are tested. The monthly RBI panel was assembled by hand and is not rebuilt by the project code.
12. **Market frictions.** The analysis uses FBIL benchmark fixings rather than traded prices, and ignores bid–ask spreads and the timing difference between the Mumbai fixing and US closing rates.

6. Conclusion

For short-dated USD/INR forwards, most of the day-to-day variance in hedging cost comes from the basis rather than from interest rates. The basis has a clear term structure and a financial year-end charge that can be measured daily and drops out at the rollover. RBI dollar purchases are followed by a flatter basis curve. Standard VaR models that ignore this structure, RiskMetrics and historical simulation, understated short-dated basis risk in 2024–26. A filtered historical simulation built on the state-space forecast did better over the same period, but that result needs confirmation on data after October 2026 and on actual forward books.

Appendix A. Basis construction

For a contract with n days from spot to forward settlement, rupee rate i_INR and dollar rate i_USD (both in percent, for n days), the CIP-implied premium in annualised basis points is

$$\text{implied} = \left[\frac{1 + i_{\text{INR}} \cdot \frac{n}{365}}{1 + i_{\text{USD}} \cdot \frac{n}{365}} - 1\right] \times \frac{365}{n} \times 10{,}000$$

and the basis is the FBIL annualised premium minus this number. Rates for each n are obtained by linear interpolation in days between curve points. The basis in bp of notional, used for the FBD April contract, is the annualised basis multiplied by n/365.

Rate checks (Table A1) compare the MIBOR-OIS curve with the Treasury bill curve and the US constant-maturity yields with traded bill rates. The OIS–bill gap averages between −7 and +18 bp across tenors. The US bill–CMT gap is under 9 bp at 1M and close to zero at 6M and 1Y.

Table A1. Interest-rate checks (percentage points)

Comparison	Days	Mean	SD
INR OIS - T-bill 1M	1977	0.178	0.210
INR OIS - T-bill 3M	1977	0.063	0.195
INR OIS - T-bill 6M	1977	-0.064	0.206
INR OIS - T-bill 12M	1977	-0.069	0.240
US bill 28d - CMT 30d	693	-0.087	0.038
US bill 91d - CMT 91d	693	-0.053	0.033
US bill 182d - CMT 182d	693	-0.004	0.028
US bill 364d - CMT 365d	693	0.006	0.014

Appendix B. Model specification

Let b_{it} be the basis at tenor i on day t, with maturity τ_i in months and n_{it} days. The observation equation is:

$$b_{it} = L_t + g(\tau_i)S_t + h(\tau_i)C_t + x_{it}D_t + u_{it}$$

where the loading functions are defined as:

$$g(\tau) = \frac{1 - e^{-\lambda\tau}}{\lambda\tau}, \quad h(\tau) = g(\tau) - e^{-\lambda\tau}, \quad \lambda = 0.30$$

and the year-end indicator x_{it} is defined as:

$$x_{it} = \begin{cases} \frac{365}{n_{it}} & \text{if the contract settles on or after the first business day of April after the coming 31 March} \\ 0 & \text{otherwise} \end{cases}$$

This converts the charge D_t, which is in bp of notional, into an annualised rate for the contract. The First Business Day (FBD) April contract, observed in bp of notional with n_t days, gives a further equation:

$$f_t = \frac{n_t}{365}[L_t + g(\tau_t)S_t + h(\tau_t)C_t] + D_t + u_{ft}, \quad \tau_t = \frac{n_t}{365/12}$$

The curve factors $F_t = (L_t, S_t, C_t)'$ follow a VAR(1) with diagonal persistence:

$$F_t = (I - \Phi)\mu + \Phi F_{t-1} + \eta_t, \quad \eta_t \sim \mathcal{N}(0, \Sigma), \quad \Sigma = \Lambda\Lambda'$$

The year-end charge is a random walk within each financial year and restarts after each rollover:

$$\begin{aligned} D_t &= D_{t-1} + w_t, \quad w_t \sim \mathcal{N}(0, \sigma_w^2) \quad \text{within a season;} \\ D_t &= \mu_D + \xi_t, \quad \xi_t \sim \mathcal{N}\left(0, \sigma_\xi^2\right) \quad \text{at the restart} \end{aligned}$$

The tenor deviations are AR(1), with innovations correlated across tenors according to their distance in months:

$$u_{it} = \rho u_{i,t-1} + e_{it}, \quad \mathrm{Cov}\left(e_{it}, e_{jt}\right) = \sigma_i \sigma_j \kappa^{|m_i - m_j|}(1 - \rho^2)$$

The FBD deviation u_{ft} has the same form with scale σ_{fbd}. There is no separate measurement error: all noise is carried in the deviation states. The model has 17 states (three curve factors, the year-end charge, twelve tenor deviations and the FBD deviation) and 30 parameters. Persistence and correlation parameters are constrained to $(-1,1)$ and variances to be positive by transformation.
Estimation is by exact maximum likelihood with the Kalman filter (Durbin and Koopman, 2012) as implemented in statsmodels (Seabold and Perktold, 2010), with approximate diffuse initialisation. Starting values come from two-step Diebold–Li estimates. The optimiser (L-BFGS) is run from two routes, the second holding the factor persistences at 0.97 before freeing them, and the higher likelihood is kept. Term-structure likelihoods can have several local maxima (Duffee and Stanton, 2012), which is why the second route is used. The comparison models in Table 5 restrict σ_w to near zero (fixed charge within a season), drop the FBD observation, or change λ.

Appendix C. Estimates and Monte Carlo study

Table C1. Parameter estimates, main model

Parameter	Estimate	Std. error	Parameter	Estimate	Std. error
phi_L	0.9741	0.0045	sigma_4M	4.7063	0.1375
phi_S	0.9765	0.0065	sigma_5M	4.7416	0.1322
phi_C	0.9749	0.0061	sigma_6M	4.6521	0.1217
mu_L	26.5281	12.4798	sigma_7M	4.5743	0.1156
mu_S	-18.0519	21.8026	sigma_8M	4.0512	0.1071
mu_C	63.5317	33.9509	sigma_9M	4.0364	0.0968
chol_00	11.5945	0.7010	sigma_10M	2.7605	0.0780
chol_10	-6.1327	1.0724	sigma_11M	2.3029	0.1046
chol_11	17.4257	0.6088	sigma_12M	4.5092	0.1047
chol_20	-21.2639	2.9910	sigma_fbd	3.6497	0.1031
chol_21	-10.4546	1.4574	mu_D	0.1227	1.2967

Parameter	Estimate	Std. error	Parameter	Estimate	Std. error
chol_22	19.1080	0.9589	sigma_w	2.6584	0.0909
sigma_1M	17.1923	0.4694	sigma_xi	0.0000	–
sigma_2M	8.2861	0.2949	rho	0.5807	0.0076
sigma_3M	6.2302	0.1762	kappa	0.4717	0.0161

chol_ij are elements of the Cholesky factor Λ of the factor shock covariance. The standard error for sigma_xi is not reported because the estimate is at zero, where the usual approximation does not apply.

The Monte Carlo study simulates ten data sets of the same length as the training sample from the estimated model and re-estimates the model on each. A second set of ten simulations sets the year-end charge to zero and provides the null distribution for the LR test. Of the 20 fits, 19 converged. Table C2 shows parameter recovery for the main parameters and Table C3 recovery of the states.

Table C2. Monte Carlo parameter recovery (10 replications)

Parameter	True	Mean	SD	Min	Max
phi_L	0.974	0.975	0.009	0.962	0.992
phi_S	0.977	0.982	0.015	0.962	0.999
phi_C	0.975	0.973	0.014	0.959	0.999
mu_L	26.528	22.970	8.141	7.698	37.735
mu_S	-18.052	-11.095	16.831	-37.802	9.524
mu_C	63.532	67.440	26.753	23.333	115.390
sigma_1M	17.192	17.059	0.542	16.357	18.367
sigma_3M	6.230	6.211	0.176	5.955	6.456
sigma_12M	4.509	4.337	0.243	3.941	4.648
sigma_fbd	3.650	3.644	0.118	3.469	3.813
mu_D	0.123	-0.751	1.421	-3.027	1.777
sigma_w	2.658	2.658	0.091	2.501	2.800
rho	0.581	0.581	0.008	0.569	0.598
kappa	0.472	0.464	0.024	0.438	0.501

The year-end parameters (σ_w, ρ, κ, σ_fbd) and the deviation scales are recovered closely. The long-run means, the slope and curvature persistences and μ_D are not. Persistence reached the 0.999 bound in 4 of 10 replications.

Table C3. Monte Carlo recovery of the states (average over replications)

State	Estimate	RMSE (bp)	Bias (bp)	90% band coverage
curvature	filtered	52.81	0.59	0.882
curvature	smoothed	45.01	0.88	0.890
level	filtered	14.98	-0.18	0.871
level	smoothed	13.02	-0.21	0.877
slope	filtered	15.32	0.02	0.882
slope	smoothed	13.99	-0.03	0.884
turn	filtered	2.27	0.01	0.892
turn	smoothed	1.99	-0.01	0.895

Appendix D. Risk models and tests

All forecasts are for the one-day change in the basis at a constant tenor. Upper-tail VaR is reported, since a rise in the basis raises the cost of buying dollars forward.

- **HS:** empirical quantile of the past 250 daily changes.
- **RiskMetrics:** zero mean, EWMA variance with $\lambda = 0.94$, normal quantile (J.P. Morgan/Reuters, 1996).
- **EWMA-t:** EWMA variance with Student-t quantile; the mean adds the average training-period year-end charge when a contract starts or stops spanning the year end; degrees of freedom fitted on the training sample.
- **SS-normal:** mean and variance of the state-space one-step forecast, normal quantile.
- **SS-t:** state-space mean; the state-space standard deviation is rescaled by an EWMA of squared standardised errors, with the weight chosen per tenor on the training sample; Student-t quantile.
- **SS-FHS:** same mean and scale as SS-t; quantile and ES taken from the empirical distribution of all past standardised errors (at least 250), in the spirit of Barone-Adesi, Giannopoulos and Vosper (1999) and Hull and White (1998).

Tests: the Kupiec (1995) test of the breach rate; the Christoffersen (1998) test of independence of breaches; the Basel traffic light (BCBS, 1996) with cut-offs rescaled to the test length; the Du and Escanciano (2017) unconditional ES test; and Diebold and Mariano (1995) tests of the difference in tick loss (VaR) and FZ0 loss (VaR and ES jointly; Fissler and Ziegel, 2016; Patton, Ziegel and Chen, 2019). Days on which a model's ES is not positive are dropped from the FZ0 comparison for that pair (at most four days).

Table D1. Number of 95% VaR breaches in the test period (Basel-style zone)

Tenor	HS	RiskMetrics	EWMA-t	SS-normal	SS-t	SS-FHS
1M	46 (Y)	42 (G)	65 (R)	32 (G)	63 (R)	43 (Y)
2M	52 (Y)	43 (Y)	57 (R)	26 (G)	61 (R)	46 (Y)
3M	48 (Y)	38 (G)	51 (Y)	26 (G)	66 (R)	40 (G)
4M	48 (Y)	42 (G)	47 (Y)	33 (G)	55 (Y)	41 (G)
5M	51 (Y)	36 (G)	41 (G)	30 (G)	46 (Y)	41 (G)
6M	50 (Y)	41 (G)	40 (G)	25 (G)	42 (G)	36 (G)
7M	48 (Y)	43 (Y)	44 (Y)	24 (G)	46 (Y)	35 (G)
8M	48 (Y)	39 (G)	42 (G)	18 (G)	40 (G)	38 (G)
9M	46 (Y)	38 (G)	45 (Y)	17 (G)	46 (Y)	41 (G)
10M	46 (Y)	39 (G)	45 (Y)	17 (G)	44 (Y)	41 (G)
11M	58 (R)	36 (G)	47 (Y)	14 (G)	43 (Y)	36 (G)
12M	51 (Y)	38 (G)	49 (Y)	16 (G)	48 (Y)	41 (G)

Expected breaches: 33.3.

Table D2. Du–Escanciano ES test p-values, 99% level

Tenor	HS	RiskMetrics	EWMA-t	SS-normal	SS-t	SS-FHS
1M	<0.001	<0.001	<0.001	<0.001	<0.001	0.950
2M	<0.001	<0.001	<0.001	<0.001	<0.001	0.243
3M	0.072	<0.001	<0.001	<0.001	<0.001	0.311
4M	<0.001	<0.001	0.003	<0.001	<0.001	0.104
5M	<0.001	<0.001	<0.001	0.003	<0.001	0.003
6M	<0.001	<0.001	<0.001	<0.001	<0.001	0.171
7M	<0.001	<0.001	0.005	0.144	0.001	0.009
8M	<0.001	<0.001	0.004	0.051	0.001	0.365
9M	<0.001	<0.001	0.002	0.067	0.016	0.105
10M	<0.001	<0.001	<0.001	0.043	0.003	0.102
11M	0.006	<0.001	0.003	0.089	0.039	0.245
12M	0.006	<0.001	0.418	0.795	0.193	0.133

Table D3. SS-FHS against benchmarks, 99% level: Diebold–Mariano p-values

Tenor	vs RiskMetrics: tick	vs RiskMetrics: FZ0	vs HS: tick	vs HS: FZ0	vs EWMA-t: tick	vs EWMA-t: FZ0
1M	0.044	0.019	0.071	0.002	0.049	0.077
2M	0.078	0.038	0.258	0.012	0.124	0.300
3M	0.079	0.026	0.267	0.054	0.100	0.047
4M	0.110	0.027	0.425	0.121	0.295	0.304
5M	0.064	0.015	0.271	0.080	0.097	0.165
6M	0.039	0.037	0.117	0.065	0.030	0.071
7M	0.270	0.084	0.367	0.244	0.196	0.610
8M	0.214	0.214	0.368	0.365	0.073	0.236
9M	0.561	0.111	0.751	0.391	0.233	0.313
10M	0.146	0.069	0.403	0.267	0.051	0.027
11M	0.149	0.058	0.294	0.160	0.070	0.059
12M	0.150	0.248	0.259	0.276	0.280	0.826

The loss difference favours SS-FHS in every comparison except FZ0 against EWMA-t at 7M; the p-values show whether the difference is significant.

Table D4. Chosen EWMA weights and degrees of freedom for SS-t (training sample)

Tenor	EWMA weight	Degrees of freedom
1M	0.80	3.79
2M	0.85	4.16
3M	0.85	3.90
4M	0.90	4.29
5M	0.90	4.40
6M	0.90	5.87

Tenor	EWMA weight	Degrees of freedom
7M	0.90	4.77
8M	0.90	4.56
9M	0.94	3.88
10M	0.90	4.19
11M	0.90	4.14
12M	0.90	4.19

Appendix E. Local projection specification

For factor y (monthly average of the smoothed level or slope) and driver x, the regression at horizon h is

$$y_{t+h} = \alpha_h + \beta_h x_t + \sum_{j=1}^{p} \gamma_j{}' z_{t-j} + \text{month dummies} + \varepsilon_{t+h}$$

where z contains the factor and all four drivers: net spot purchases (USD billion), the change in RBI's net forward position (USD billion), the change in net liquidity injection (₹ lakh crore) and the log change in the broad dollar index. The lag length p is chosen by AIC in a VAR on the same variables, with a minimum of two (two was chosen). β_h is the response. Standard errors are Newey–West with h + 1 lags (Newey and West, 1987); 90% bands are reported. Months with fewer than ten trading days are excluded.